Final project report

Our HVAC system was designed to oversee the temperatures of multiple rooms and regulate them to their desired temperatures using dampers, fans, and a heater and cooler. This is important because the temperatures of rooms will naturally change throughout the day so it's important for the system to be aware of these changes and edit the system configuration accordingly. This system has two modes, manual and automatic.

In manual mode you can trigger all these components individually to reach a desired airflow, or better yet you can set each of the rooms to a set point and in automatic mode the system will detect which rooms need to be heated or cooled and adjust the system configuration accordingly.

Our logic works by passing a variable that determines switch room, if any, is currently not at the desired set point. Once a room that is not at its set point is detected the variable is changed to identify this room as the room currently being worked on and will edit the HVAC system configution to either heat or cool just this room. After this, our system will continue through the primary node logic loop until it loops again and returns to this point in the code. Once it's at this point it will again check if the temperature is at the desired setpoint and if it isn't it will continue through the primary node loop again, and if it is it will stop the heating or cooling and open up the dampers to all of the rooms again before continuing through the primary node loop once again checking if any of the rooms are different from their desired setpoints and repeat the process.

In the digital age, our HVAC system is at potential risk of threats that could compromise our system. The security of our system must be updated and properly assessed to ensure that no one or thing can breach it with malicious intent. When it comes to eavesdropping, it isn't that big of a threat. Our system sends data about heating/cooling, damper positions, and room temperatures. If someone were to get this info without us knowing they couldn't do anything about it.

Wi-Fi connection is essential for our system to operate correctly with the node-red dashboard. Jamming of our network could be detrimental to our system due to it using MQTT to transfer all data. If our MQTT were compromised, our system would fail. If we relied partially on sockets to send data, it could help counter a compromise in our network or MQTT brokers. If someone were to hack into our primary control node, they would have access to controlling our damper positions and commands over heat/cool. Still having access to the secondary control node could help counter act their controls by turning heat/cool off completely. The same can be done if it was the other way around and the secondary control node was under someone's control. If someone were to impersonate

our temperature values, it would trick our automatic system into heating/cooling based off of false values. This could potentially be dangerous for people in the building if extreme temperatures are being displayed.

Through each lab, our HVAC system as gone through many iterations. The final version was able to maintain three separate room temperatures and regulate their designated dampers. Although the logic behind deciding which room needs heating/cooling could be revised, our latency with dampers was surprisingly better then other systems.